COVID-19 Global Impact Dashboard

Executive Summary — Veda Technology Major Project

This project builds a reproducible analytics pipeline for country/date COVID-19 data. The analysis emphasizes population-normalized cases and deaths, 7-day rolling averages, vaccination rollout, case fatality rate, doubling time, and wave detection.

Area	Approach
Data	Our World in Data COVID-19 compact dataset; WHO and official-source inputs.
Cleaning	Date standardization, numeric coercion, duplicate removal, conservative missing-value handling.
Derived metrics	7-day rolling averages, cases/deaths per million, crude CFR, 7-day growth-based doubling time.
Visualization	Country trend lines, population-normalized comparisons, vaccination rollout and wave peaks.
Interpretation	Descriptive comparison only; vaccination/death relationship is treated as association, not causation.

Key sourced observations from the included preview snapshot

On 12 Aug 2024, OWID reported at least one vaccine dose for 72.1% of India's population, 79.1% of the United States, 90.2% of Brazil and 94.3% of Portugal. Completed initial-protocol coverage was 66.8% in India, 67.5% in the United States, 83.8% in Brazil and 85.5% in Portugal. The spread shows why vaccination rollout should be compared by population share rather than raw dose counts.

The full notebook downloads the current OWID dataset and should be run before final submission if exact current values are required. Static preview charts are separated from the reproducible time-series analysis.

Source: Our World in Data COVID-19 documentation and data pages. See README.md for the source links and attribution.